

2.28 $\forall x \in [-1, 1], f_n(x) = 0$ for n large enough, so $\lim_{n \rightarrow \infty} f_n(x) = f(x) = 0$. Also, $\int f_n d\mu = 1 - 1 = 0 = \int f d\mu$. For any $t > 0$, $\sup_{n \geq 1} \int_{|f_n| > t} |f_n| d\mu = 2$, so $\{f_n\}_{n \geq 1}$ is not UI.

2.30 $|f_n| \leq n^{-1/2} \leq 1 \Rightarrow \lim_{n \rightarrow \infty} f_n(x) = 0$, for all $x \in \mathbb{R}$. And $\forall t > 1, \int |f_n| I\{|f_n| > t\} d\mu = 0 \Rightarrow \{f_n\}_{n \geq 1}$ is UI. However, $\int f_n d\mu = \sqrt{n} \rightarrow \infty$ as $n \rightarrow \infty$. So $\int f_n d\mu \not\rightarrow \int f d\mu$.

(It's more important to understand the purpose of those counter examples, see **Remark 2.5.4** for details.)

2.36 (a) $\forall \epsilon > 0, \lim_{n \rightarrow \infty} \int_{\Omega} |f_n| d\mu = 0 \Rightarrow \exists n_{\epsilon} \in \mathbb{N}$, such that $\sup_{n \geq n_{\epsilon}} \int_{\Omega} |f_n| d\mu < \epsilon$. Moreover, by **DCT**, $\exists t_{\epsilon} > 0$, such that $\sup_{1 \leq n \leq n_{\epsilon}} \int |f_n| I\{|f_n| > t\} d\mu < \epsilon$ for all $t > t_{\epsilon}$. That's to say, $\forall \epsilon > 0, \exists t_{\epsilon} > 0$ such that for all $t > t_{\epsilon}$,

$$\sup_{n \geq 1} \int |f_n| I\{|f_n| > t\} d\mu \leq \max \left(\sup_{1 \leq n \leq n_{\epsilon}} \int |f_n| I\{|f_n| > t\} d\mu, \sup_{n \geq n_{\epsilon}} \int_{\Omega} |f_n| d\mu \right) \leq \epsilon.$$

So $\{f_n\}_{n \geq 1}$ is UI.

(b) First, let's prove an improved version of **Proposition 2.5.7** (iv) by simply mimicking the proof there. The statement is:

For any $p > 0$, if $\{|f_{\lambda}|^p : \lambda \in \Lambda\}$ and $\{|g_{\gamma}|^p : \gamma \in \Gamma\}$ are UI on $(\Omega, \mathcal{F}, \mu)$, then so is $\{|f_{\lambda} + g_{\gamma}|^p : \lambda \in \Lambda, \gamma \in \Gamma\}$.

Let $h_t(x) = xI_{(t, \infty)}(x)$, $a(t) = \sup_{\lambda \in \Lambda} \int_{\Omega} h_t(|f_{\lambda}|^p) d\mu$, $b(t) = \sup_{\gamma \in \Gamma} \int_{\Omega} h_t(|g_{\gamma}|^p) d\mu$. Then

$$\int h_t(|f_{\lambda} + g_{\gamma}|^p) d\mu \leq \int h_t[\{2 \max(|f_{\lambda}|, |g_{\gamma}|)\}^p] d\mu \leq \int h_t(2^p |f_{\lambda}|^p) d\mu + \int h_t(2^p |g_{\gamma}|^p) d\mu \leq a(t/2^p) + b(t/2^p).$$

So $\{|f_{\lambda} + g_{\gamma}|^p : \lambda \in \Lambda, \gamma \in \Gamma\}$ is UI (UI is a concept built on integrable functions, so when we say a class of functions is UI, it automatically implies integrability). This result convinced us that $\{|f_n - f|^p\}_{n \geq 1}$ is UI.

Secondly, we are going to show that $f \in L^p(\Omega, \mathcal{F}, \mu)$. Because $f_n \xrightarrow{m} f$, there exists a sub-sequence $\{f_{n_k}\}_{k \geq 1}$ such that $f_{n_k} \rightarrow f$ a.e. (μ) , which also means $|f_{n_k}|^p \rightarrow |f|^p$ a.e. (μ) . Together with $\mu(\Omega) < \infty$ and $\{|f_n|^p\}_{n \geq 1}$ is UI, we get $|f|^p \in L^1(\mu)$, thus $f \in L^p(\mu)$.

Now, for any $\epsilon, t > 0$

$$\begin{aligned} & \int |f_n - f|^p d\mu \\ &= \int |f_n - f|^p I\{|f_n - f|^p \leq \epsilon\} d\mu + \int |f_n - f|^p I\{\epsilon < |f_n - f|^p \leq t\} d\mu + \int |f_n - f|^p I\{|f_n - f|^p > t\} d\mu \\ &\leq \epsilon^p \mu(\Omega) + t^p \mu(\{\omega : |f_n(\omega) - f(\omega)| > \epsilon\}) + \sup_{n \geq 1} \int h_t(|f_n - f|^p) d\mu \end{aligned}$$

Take $\limsup_{n \rightarrow \infty}$ on both sides of the above inequality, we get

$$\limsup_{n \rightarrow \infty} \int |f_n - f|^p d\mu \leq \epsilon^p \mu(\Omega) + \sup_{n \geq 1} \int h_t(|f_n - f|^p) d\mu$$

for any $\epsilon, t > 0$. Let $\epsilon \rightarrow 0, t \rightarrow \infty$, by the UI of $\{|f_n - f|^p\}_{n \geq 1}$ and $\mu(\Omega) < \infty$, $\limsup_{n \rightarrow \infty} \int |f_n - f|^p d\mu = 0$.

2.42 To better illustrate our problem, I consider the following diagram

$$(\Omega_1, \mathcal{F}_1, \mu_1) \xrightarrow{f} (\Omega_2, \mathcal{F}_2, \mu_2) \xrightarrow{h} (\mathbb{R}, \mathcal{B}(\mathbb{R}), \mu_3) \xrightarrow{I(x) \equiv x} (\mathbb{R}, \mathcal{B}(\mathbb{R}), \mu_4),$$

where $\mu_2 \equiv \mu_1 \circ f^{-1}$, $\mu_3 \equiv \mu_2 \circ h^{-1}$, $\mu_4 = \mu_3 \circ I^{-1} = \mu_3$. $g \equiv h \circ f$. We want to show

$$g \in L^1(\mu_1) \Leftrightarrow h \in L^1(\mu_2) \text{ and } \int_{\Omega_1} g d\mu_1 = \int_{\Omega_2} h d\mu_2. \quad (*)$$

i. For h being any indicator function I_A , $A \in \mathcal{F}_2$,

$$I_A \in L^1(\mu_2) \Leftrightarrow \int_{\Omega_2} I_A(x) d\mu_2(x) < \infty \Leftrightarrow \mu_2(A) = \mu_1 \circ f^{-1}(A) < \infty \Leftrightarrow \int_{\Omega_1} I_{f^{-1}(A)}(\omega) d\mu_1(\omega) < \infty \Leftrightarrow I_A \circ f \in L^1(\mu_1)$$

Clearly,

$$\int_{\Omega} I_A \circ f(\omega) d\mu_1(\omega) = \int_{\Omega_1} I_{f^{-1}(A)}(\omega) d\mu_1(\omega) = \mu_1(f^{-1}(A)) = \mu_1 \circ f^{-1}(A) = \mu_2(A) = \int_{\Omega_2} I_A(x) d\mu_2(x).$$

So $(*)$ is true for h being indicator function, which implies that $(*)$ is true for simple nonnegative measurable functions, who can always be written as finite linear combination of indicator functions with disjoint indicator sets.

ii. For h being any nonnegative measurable function, there always exists an increasing sequence of nonnegative measurable functions $h_n, n = 1, 2, \dots$, such that $\lim_{n \rightarrow \infty} h_n = h$. By **MCT** and the result we proved in part i.,

$$\int_{\Omega_2} h d\mu_2 = \lim_{n \rightarrow \infty} \int_{\Omega_2} h_n d\mu_2 = \lim_{n \rightarrow \infty} \int_{\Omega_1} h_n \circ f d\mu_1 = \int_{\Omega_1} h \circ f d\mu_1 = \int_{\Omega_1} g d\mu_1.$$

Thus $(*)$ holds for h being any nonnegative measurable function.

iii. In general, for any measurable function h , using the results proved above,

$$h \in L^1(\Omega_2) \Leftrightarrow h^+, h^- \in L^1(\Omega_2) \Leftrightarrow h^+ \circ f, h^- \circ f \in L^1(\Omega_1) \Leftrightarrow h \circ f = g \in L^1(\Omega_1).$$

And when $h \in L^1(\Omega_2)$,

$$\int_{\Omega_2} h d\mu_2 = \int_{\Omega_2} h^+ d\mu_2 - \int_{\Omega_2} h^- d\mu_2 = \int_{\Omega_1} h^+ \circ f d\mu_1 - \int_{\Omega_1} h^- \circ f d\mu_1 = \int_{\Omega_1} (h^+ - h^-) \circ f d\mu_1 = \int_{\Omega_1} g d\mu_1.$$

Thus $(*)$ is truly true.

The other part of this problem is a direct easy application of the result we proved above. Think about $\tilde{h} = I(x)$ and $\tilde{f} = h, \tilde{g} = \tilde{h} \circ \tilde{f} = I \circ h = h, \tilde{\mu}_1 = \mu_3, \tilde{\mu}_2 = \mu_4$, the proof is then finished by applying the above result.